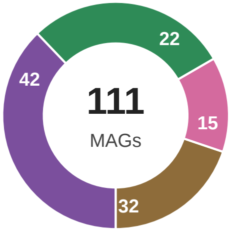
Livestock-waste metagenomes yield two convergent lineages with a vertically inherited *ure–cah* module

a genomic chassis for alkali-tolerant microbially induced carbonate precipitation (MICP)

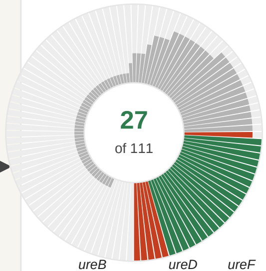
111 MAGs from four waste streams



- cattle n = 22
- swine n = 15
- sheep n = 32
- poultry n = 42

cattle · swine · sheep · poultry slurries

MICP module score across the panel



- MICP-complete, n = 6
- *ureA–G* + *cah*, n = 27
- score 0–7

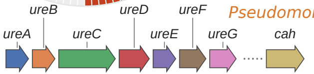
bar length = genes present

centre = MAGs with all eight

two convergent lineages

Sphingobacterium × 4

Pseudomonas_E × 2



0/6 *ure* contigs on a mobile element

vertically inherited in both lineages

What the six MICP-complete MAGs share

42/42 — UreC active-site residues identical to *S. pasteurii*

11.7× — Mrp Na^+/H^+ antiporter prevalence — the alkali-tolerance signature

3.07 % — of 7,599 livestock species clusters carry the module

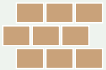
S13 · S16 — novel *Sphingobacterium* — ANI 94.6 / 93.8 % to RefSeq

urease fold kept 6/6 · ESMFold TM > 0.5

waste-coupled MICP

urea $\xrightarrow{\text{urease}}$ NH_4^+ + CO_2 $\xrightarrow{\text{carbonic anhydrase}}$ CO_3^{2-} + Ca^{2+} $\xrightarrow{\text{pH ↑ · nucleation}}$

$CaCO_3$ →



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